

1. INNER JOIN

Retrieve employees along with their department name and the projects they work on

```
1 SELECT E.Fname, E.Lname, D.DName, P.PName
2 FROM Employee E
3 INNER JOIN Department D
4     ON E.DNUM = D.DNUM
5 INNER JOIN Works_On W
6     ON E.SSN = W.SSN
7 INNER JOIN Project P
8     ON W.PNumber = P.PNumber;
```

Query Result x

SQL | All Rows Fetched: 4 in 0.47 seconds

	FNAME	LNAME	DNAME	PNAME
1	Ali	Al-Khan	Information Tech	Website
2	Sara	Ahmed	HR	Payroll
3	Omar	Hassan	Finance	Bank System
4	Noura	Ali	Sales	CRM

2. LEFT JOIN

Display all employees with their department information and dependent details, including employees who have no

```
1 SELECT E.Fname, E.Lname, D.DName, DP.DependentName, DP.Gender
2 FROM Employee E
3 LEFT JOIN Department D
4     ON E.DNUM = D.DNUM
5 LEFT JOIN Dependent DP
6     ON E.SSN = DP.SSN;
```

Query Result x

SQL | All Rows Fetched: 5 in 0.036 seconds

	FNAME	LNAME	DNAME	DEPENDENTNAME	GENDER
1	Ali	Al-Khan	Information Tech	Aisha	F
2	Sara	Ahmed	HR	Zaid	M
3	Omar	Hassan	Finance	Mona	F
4	Noura	Ali	Sales	(null)	(null)
5	Khalid	Salim	Security	Lama	F

3. RIGHT JOIN

Show all projects with their assigned department and the employees working on them, including projects with no assigned employees

Worksheet

Query Builder

```
1 SELECT P.PName, D.DName, E.Fname, E.Lname
2 FROM Employee E
3 RIGHT JOIN Works_On W
4     ON E.SSN = W.SSN
5 RIGHT JOIN Project P
6     ON W.PNumber = P.PNumber
```

Query Result x

    SQL | All Rows Fetched: 5 in 0.019 seconds

	PNAME	DNAME	FNAME	LNAME
1	Website	Information Tech	Ali	Al-Khan
2	Payroll	HR	Sara	Ahmed
3	Bank System	Finance	Omar	Hassan
4	CRM	Sales	Noura	Ali
5	Security App	Security	(null)	(null)

4. SELF JOIN

Retrieve each employee along with their supervisor's name and department

Worksheet

Query Builder

```
1 SELECT E.Fname AS Employee_First_Name,
2     E.Lname AS Employee_Last_Name,
3     S.Fname AS Supervisor_First_Name,
4     S.Lname AS Supervisor_Last_Name,
5     D.DName
6 FROM Employee E
7 LEFT JOIN Employee S
8     ON E.Supervisor = S.SSN
9 LEFT JOIN Department D
10    ON E.DNUM = D.DNUM;
```

Query Result x

    SQL | All Rows Fetched: 5 in 0.011 seconds

	EMPLOYEE_FIRST_NAME	EMPLOYEE_LAST_NAME	SUPERVISOR_FIRST_NAME	SUPERVISOR_LAST_NAME	DNAME
1	Ali	Al-Khan	(null)	(null)	Information Tech
2	Sara	Ahmed	Ali	Al-Khan	HR
3	Omar	Hassan	Ali	Al-Khan	Finance
4	Noura	Ali	Sara	Ahmed	Sales
5	Khalid	Salim	Omar	Hassan	Security